

Week 2 Report — Problem Finalization, Research & Market Analysis

Programme: IIMSTC – IoT & Embedded Systems (Cohort 4)
Team Size: 15 Members

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Day 1 — Problem Finalization

The team finalized the problem statement as **IoT-Driven Cold Chain System for Real-Time Monitoring and Predictive Failure Prevention**. This decision was made after evaluating feasibility, industry relevance, and potential impact. The problem was clearly defined in terms of objectives, scope, and expected outcomes to establish a strong direction for further development.

Day 2 — Literature Survey & System Understanding

A comprehensive literature survey was conducted by reviewing research papers, journals, and technical articles related to cold chain monitoring systems and IoT-based solutions. The team gained a clear understanding of system requirements such as temperature and humidity monitoring, real-time data transmission, alert mechanisms, and predictive maintenance. Various application domains, including pharmaceuticals and food logistics, were also explored.

Day 3 — Research on Existing Technologies

Each team member carried out individual research on existing technologies used in cold chain systems. This included studying IoT sensors, cloud platforms, GPS tracking systems, and data analytics tools. The architecture and working principles of current solutions were analyzed to understand commonly used components and system designs.

Day 4 — Gap Analysis & Innovation Opportunities

Based on the literature survey and technical research, several limitations in existing systems were identified. These included lack of predictive failure capabilities, dependency on continuous

connectivity, high implementation costs, and limited integration between monitoring and response mechanisms. These insights led to the identification of potential innovation areas such as predictive analytics, cost-effective design, and improved system reliability.

Day 5 — Cost & Feasibility Analysis

The team performed a detailed cost and feasibility analysis of the proposed system. This involved estimating the cost of hardware components such as sensors, microcontrollers, and communication modules, along with cloud infrastructure requirements. Feasibility was evaluated in terms of scalability, maintainability, and real-world deployment challenges.

Day 6 — Market Survey & Industry Insights

A structured market survey was conducted to understand industry needs, user expectations, and challenges in existing cold chain systems. The team analyzed different application areas, pricing models, and customer pain points such as lack of real-time monitoring, product spoilage, and inefficient alert systems. These insights helped refine the proposed solution to better align with real-world requirements.

Summary

Week 2 focused on finalizing the problem statement and strengthening the project through detailed research, literature survey, and market analysis. The team successfully identified key gaps in existing systems and validated the feasibility and industry relevance of the proposed solution. This phase established a solid foundation for moving into system design and implementation in the upcoming weeks.